Suspended Sediment Particle Size Prediction

Comparing Random Forest, Support Vector Regression, and XGBoost

Problem statement

Suspended sediment particle-size distributions influence transport and aquatic processes. This project compares three regression models for estimating median particle diameter (d50) and particle-size-distribution variance (sigma2) from environmental measurements. The work is a reproducible educational comparison, not a field-validated prediction system.

Dataset details

Source and nature. The supplied file, synthetic_data_improved.csv, is a synthetic hourly dataset. It contains 1,648 records, eight environmental predictors (salinity S, near-bottom velocity ub, refractive index np, water temperature T, two optical ratios, chlorophyll-a chl_a, and mean tidal velocity u), and the two target variables d50 and sigma2. Predictor columns contain missing values; both targets are present. The CSV is not verified as the original Stanford dataset, nor as measured field data.

Preparation. Records were sorted by timestamp. The timestamp was used to preserve order for splitting and was excluded from model inputs. The predictor matrix was imputed with the training-set median; the final 20% of observations formed the chronological test set.

Approach and implementation

Separate models were fitted for d50 and sigma2. Random Forest provided a bagged-tree baseline; SVR provided a scaled kernel-based model; XGBoost added a gradient-boosted tree comparison. SVR inputs were standardized after imputation. Performance was evaluated on the held-out period using R^2, root mean squared error (RMSE), and mean absolute error (MAE). The project includes a Python pipeline and a Jupyter notebook to reproduce the results.

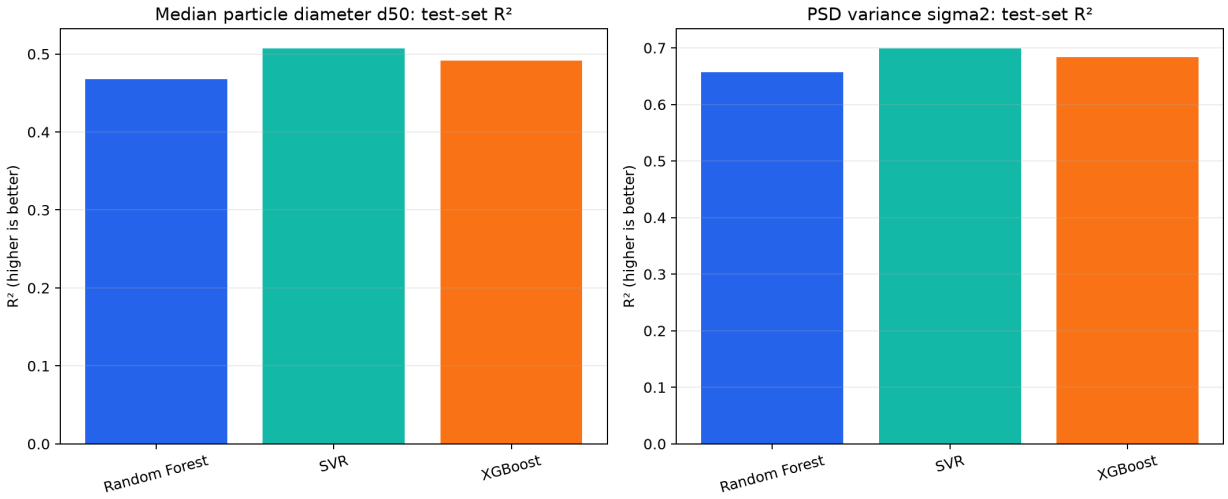
Results

Target	Model	R^2	RMSE	MAE
d50	Random Forest	0.468	43.98	16.08
d50	SVR	0.507	42.34	15.34
d50	XGBoost	0.491	43.01	15.47
sigma2	Random Forest	0.657	19.72	7.74
sigma2	SVR	0.699	18.47	7.03
sigma2	XGBoost	0.684	18.92	7.47

Across both targets, SVR achieved the highest test R^2 and the lowest RMSE and MAE in this run. The result applies only to this synthetic dataset and chronological split.

Findings and project notes

Interpretation, limitations, and reproducibility



Conclusion

On the held-out final segment of this file, SVR ranked first for both targets. For d50, it achieved $R^2 = 0.507$, RMSE = 42.34 and MAE = 15.34; for sigma2, $R^2 = 0.699$, RMSE = 18.47 and MAE = 7.03. Random Forest and XGBoost were close but lower on all three metrics. This comparison suggests SVR fits the patterns in this generated dataset best under the chosen setup; it does not establish that SVR is best for real sediment measurements.

Limitations

The dataset is synthetic and contains a small number of deliberately generated target patterns, including extreme values. Model scores can reflect those construction rules. The test segment is from the same generated series, so it is not independent validation across locations, instruments, or field campaigns. Results may change with alternative split dates, preprocessing, or hyperparameter tuning. No hyperparameter search or uncertainty analysis was performed. Obtain and cite the original observational data before making claims about real-world prediction or claiming Stanford-data reproduction.

Reproducibility

The accompanying private-repository-ready folder contains the CSV, run_project.py, requirements.txt, README.md, and notebooks/01_model_comparison.ipynb. From the project directory, install dependencies with `pip install -r requirements.txt`, then run `python run_project.py`. The pipeline produces model_metrics.csv, test_predictions.csv, and r2_comparison.png in results/. The test split is chronological (first 80% train, final 20% test; 1,319 training rows and 329 test rows after sorting).

Reference context

The project topic and baseline comparison are informed by Galen Egan, “Machine learning for predicting suspended sediment particle size distributions,” Stanford CS229 project report (2019). The supplied dataset used here is synthetic and should be treated separately from that report’s data and reported results. Stanford report.

Team members: _____ _____